Multimodal Depression Detection Using Speech and Linguistic Behaviour Analysis Across Conversational Phases

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Abstract. Depression is a prevalent mental health condition that adversely impacts an individual's communication and behaviour. Diagnosis of depression relies on the self-reporting and assessment of mental health professionals, both of which are resource-intensive processes. To automate the diagnosis of depression, a preprocessing-focused multimodal machine learning (ML) system is proposed using the DAIC-WOZ clinical interview dataset. Two modality-specific data preprocessing steps are carried out on the dataset. For the audio modality, a pre-emphasis filter, bandpass filter, and amplitude normalisation are applied. For the text modality, annotations, filler words, and text pertaining to individuals other than the participants are removed. Each interview is divided into three phases containing three consecutive interview phases. From the audio recordings, 132 audio features are extracted, including Mel-Frequency Cepstral Coefficients (MFCCs) of order 13 with delta coefficients, spectral features, pitch features, jitter, shimmer, Harmonics-to-Noise Ratio, and pause features. For the textual modality, 768-dimensional Sentence-BERT text embeddings and seven text features are extracted. Six ML classifiers are trained and tested on the dataset using single-split and 10-fold cross-validation for each phase. The results of the study indicate that the multimodal approach yields higher accuracy and F1-score values than either the audio or text modality alone. The best performance is achieved by the Random Forest classifier with an accuracy of 83.6% and F1-score of 0.71 on the single-split cross-validation method for Phase 1. The text and audio features exhibit the most discriminative values for the early phases of the interviews. The results of this study indicate that audio and text modality features preprocessed through the techniques described can be accurately classified using traditional ML methods for the automated diagnosis of depression.

Keywords: Depression Detection, Multimodal Analysis, Acoustic Feature Extraction, SBERT Embeddings, Phasewise Segmentation, DAIC-WOZ, Machine Learning, Speech Processing.

1. Introduction

Across the entire earth, the World Health Organization reports that depression impacts more than 280 million people because this health condition is a very common and heavy mental illness. Verbal communication and the way a human speaks are heavily changed by depression when a person feels constant sorrow or loses their interest in life. Experts claim that many individuals suffering from this sickness do not receive a formal medical name for their struggle because tools like the PHQ-8 and PHQ-9 questionnaires depend too much on the personal opinion of the medical professional. New computer methods which study the patterns of speaking offer a hopeful way forward even though checking mental health was previously hard to provide for all people.

1.1 Motivations

The screening of depression typically involves the completion of questionnaires and doctor consultations. The issue is, patients may not always be telling the truth and doctors can think differently. There are things such as how they say things, their length of pauses, the speed of their speech and what they say, when they speak, that can indicate if someone has depression or not. We've noticed that these are signs of depression. There are recordings of the doctors speaking with the patients as well, such as that of DAIC-WOZ, which can be used for studying these signs. There is a lot of potential, for research here. We can build a system which takes into account both sound and content, and ensure that we properly prepare the data to obtain information from the system. The use of this system could provide a better understanding of depression that is possible through depression screening and depression signs.

1.2 Objectives

The main goals of this research are: (i) to develop a pipeline for preprocessing participant speech from clinical interview recordings and cleaning up associated transcripts; (ii) to extract a large, representative collection of acoustic and linguistic features from the phase segmented data from the interviews; (iii) to test and compare the performance of multiple machine learning classifiers in both unimodal and multimodal settings; and (iv) to analyse the temporal distribution of depression cues by conversational phase.

1.3 Original Contributions

The main highlights of this study are:

- A two stage preprocessing procedure that removes interviewer speech based on ground-truth timestamps, performs spectral filtering and normalises textual transcripts to provide a platform for NLP-based analysis.
- Temporal segmentation of clinical interviews, by dividing them into three equal conversational phases, allowing modelling of depression evolution across clinical interview phases.
- Converted 132 acoustic features (including MFCCs with deltas, spectral features, jitter, shimmer, HNR, and pause statistics) with 777 text features (Sentence-BERT embeddings and linguistic markers) into a 909-dimensional multimodal feature space.
- The most powerful machine learning classifiers to detect depression from unimodal and multimodal feature sets were identified by systematically assessing six classifiers through 10-fold stratified cross validation, which revealed CatBoost and Random Forest.

1.4 Paper Layout

The rest of this paper is organised as follows. In section 2, related research work for speech-based and multimodal depression detection is reviewed. The proposed system and data set and methodologies are described in Section 3. Experimental results and evaluation are shown in Section 4. The comparative performance analysis is presented in Section 5. Section 6 elaborates on the contributions and Section 7 summarizes the contributions and future directions.

2. Literature Survey

The field of automated depression detection has made significant advances in recent ten years, with the development of multi-modality architectures and systems that rely on only a single sensory modality. Valstar et al. [1] have already conducted preliminary research that used low-level descriptor acoustic features and support vector machines on the AVEC challenge set, thus providing baseline results for depression scoring with regression methods. Later work focused on deep learning, with Cummins et al. [2] who used LSTM networks on MFCC sequences from DAIC-WOZ recordings, showing that recurrent architectures could be useful for modelling temporal dynamics of speech, but without incorporating linguistic information.

Ma et al. [3] proposed training CNNs directly from audio spectrograms, which introduced convolutional approaches. These models were able to get a competitive result, but were discarding potentially useful linguistic signals. Gong and Poellabauer [4] suggested an early fusion of ResNet audio features and text embeddings but did not implement conversational phase modelling. From the text perspective, Niu et al. [5] used BERT representations for text-based depression detection, with high F1 scores and without utilizing any acoustic features.

Yang et al. [6] investigated gated recurrent units with attention mechanisms for multimodal architectures based on transformers. These methods showed high accuracy but had high computational complexity and were not interpretable. Several works have integrated ResNet audio processing with NLP features without considering the temporal conversational dynamics [7]. Rejaibi et al. and Ray et al. analyzed LSTM and handcrafted feature methods, respectively, and both found that acoustic signals are valuable, but neither approach was able to take a phase-level temporal analysis into consideration.

One common research gap in the previous systems is the lack of phase-wise conversational analysis. The signs and symptoms of depression are not necessarily consistent throughout an interview; they might be more likely to appear in responses to early self-referential questions or later emotionally charged questions. This is addressed by the structured conversational phase segmentation, extensive preprocessing-first pipeline design and systematic integration of multimodal features that is tested on several classifiers in the present work.

Table 1. Literature survey summary of existing depression detection systems.

S. No.	**Author & Year**	**Methods/Techniques Used**	**Dataset Used**	**Research Outcomes**	**Findings**
1	Guo et al. (2022)	Feature Extraction: Transformer-based (RoBERTa, wav2vec 2.0) with Topic AttentionClassification: Two-branch multimodal transformer with late fusion	DAIC-WOZ dataset	Accuracy: 90%+ (reported improvement) Recall: Improved via attention	Effective use of topic-level attention improves detection; handles data scarcity using pre-trained models but still limited by small dataset size
2	Gan et al. (2025)	Feature Extraction: LLM (LLaMA), Audio processingClassification: Teacher–Student multimodal fusion with attention	DAIC-WOZ dataset	F1 Score: 0.991	Very high accuracy using teacher-student architecture; strong multimodal fusion but computationally complex and data-dependent
3	Gong & Poellabauer (2017)	Feature Extraction: Topic Modeling (context-aware segmentation)Classification: Multimodal feature vector + feature selection	DAIC-WOZ dataset	Improved performance over baseline methods	Captures temporal/contextual details effectively; limited by small sample size and feature dimensionality issues
4	Nykoniuk et al. (2025)	Feature Extraction: CNN + Bi-LSTM + Self-AttentionClassification: Early and Late Fusion models	DAIC-WOZ, EDAIC-WOZ datasets	Accuracy: 86% F1 Score: 0.79	Early fusion performs better; multimodal improves accuracy but dataset imbalance affects reliability

5	Yoo & Oh (2023)	Feature Extraction: CNN (audio), Transformer (text)Classification: Tensor Fusion Network + LSTM	DAIC-WOZ dataset	Moderate improvement (no exact metrics provided)	Fusion of text and voice improves detection; lacks detailed quantitative evaluation and scalability analysis
6	Xu et al. (2025)	Feature Extraction: BERT (text), wav2vec 2.0 (audio)Classification: Bi-LSTM + multi-scale convolution + adaptive pooling	CMDC & DAIC-WOZ datasets	F1 Score: Improved significantly RMSE reduced	Strong cross-modal feature integration; handles multiple datasets but requires large training data and complex architecture
7	Park & Moon (2022)	Feature Extraction: BERT-CNN (text), CNN-BiLSTM (audio)Classification: Attention-based multimodal fusion	DAIC-WOZ dataset	Improved accuracy over single-modal models	Attention-based fusion improves accuracy; still dependent on quality of preprocessing and multimodal alignment

3. Proposed System / Model

3.1 Dataset Description

This study was performed using the DAIC-WOZ (Distress Analysis Interview Corpus – Wizard-of-Oz) which is widely used as benchmark corpus for depression detection computational research . Semi-structured clinical interviews were conducted with a virtual agent named Ellie, with a focus on getting a naturalistic response from the participant on respect to his/her emotional state, daily activities, and personal experiences. Each session is synced up and accompanied by a time-stamped transcript in CSV format that labels the speaker (changes between the interviewer and the participant). Following the label filtering with the given binary ground-truth file of the "PHQ-8", 141 unique participants were left. There were three phases of conversation, and each participant provided data for each of these phases, resulting in 423 participant-phase records. The binary label (PHQ-8 Binary) was created by dichotomizing the participants into the non-depressed (0) and depressed (1) groups. The data set is skewed, with around 70.2% of the samples being non-depressed and 29.8% being depressed. The characteristics of the data set are summarised in Table 2. The proposed system schematic layout is shown as follows: The proposed framework preprocesses the audio and transcript modalities in parallel, before extracting features in a phase-wise manner and finally classifies them. The overall system is shown in Fig. 1.

Table 2. Dataset characteristics of the DAIC-WOZ corpus used in this study.

Attribute	Details
Dataset Name	DAIC-WOZ (Distress Analysis Interview Corpus – Wizard-of-Oz)
Total Participants	142 (after label filtering from 188 audio files)
Interview Format	Semi-structured clinical interview with virtual interviewer (Ellie)
Audio Format	WAV, 16 kHz mono, participant-only segments extracted
Label Type	Binary (PHQ-8 Binary: 0 = Non-depressed, 1 = Depressed)
Class Distribution	Non-depressed: 297 samples (70.2%), Depressed: 126 samples (29.8%) across 3 phases
Phases	Phase 1 (early interview), Phase 2 (mid interview), Phase 3 (late interview) — equal-thirds split
Transcript Format	CSV with columns: speaker, value, start_time, stop_time
Modalities	Audio (acoustic features) + Text (linguistic features from transcripts)
Feature Dimensionality	Audio: 132 features; Text: 777 features (incl. SBERT); Multimodal: 909 features

3.2 Schematic Layout of the Proposed System

The proposed framework processes audio and transcript modalities through parallel preprocessing pipelines before phase-wise feature extraction and classification. Figure 1 illustrates the overall system workflow.

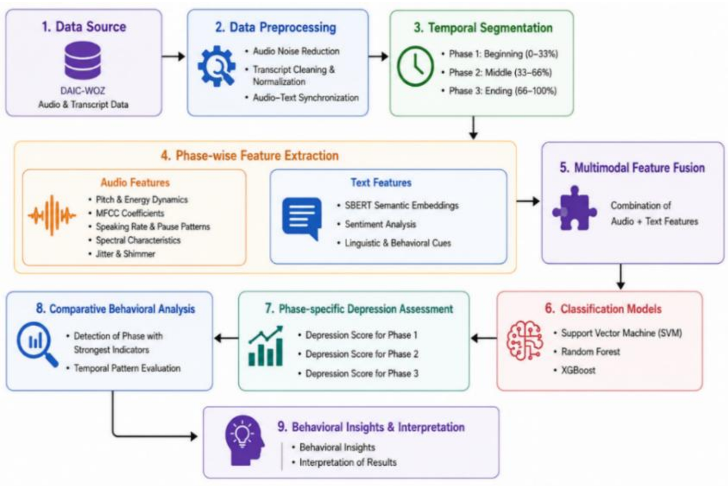

Fig. 1. Overall workflow of the proposed multimodal depression detection framework: from raw interview data through preprocessing, phase segmentation, feature extraction, fusion, and classification.

3.3 Methodologies Used

3.3.1 Audio Preprocessing.

The audio preprocessing pipeline is applied to raw DAIC-WOZ WAV files, and includes five stages of processing. The first stage involves reading the associated transcript CSV file and thus marking the boundaries of participant speech. Only segments belonging to the participant will be extracted, based on the ground truth timestamps (start_time, stop_time). This removes the virtual interviewer's voice in the absence of speaker diarisation, so that all the following extracted acoustic features are representative of the participant's behavioural speech patterns.

The participant segments extracted are then linked together in order to reconstruct the full length of the speech. In order to avoid channel imbalance artefacts, multi-channel recordings are converted to mono. The sound is re-sampled at 16000 Hz. A pre-emphasis filter ($\alpha = 0.97$) is immediately applied to enhance high frequency components that contain the important prosodic information. Lastly, a 5th order Butterworth bandpass filter (100–7500 Hz) is used to pass only frequencies in the middle band, while removing low frequencies (rumble) and high frequencies (noise), and the signal is amplitude-normalised to range $[-1, 1]$. The audio preprocessing pipeline is shown in figure 2.

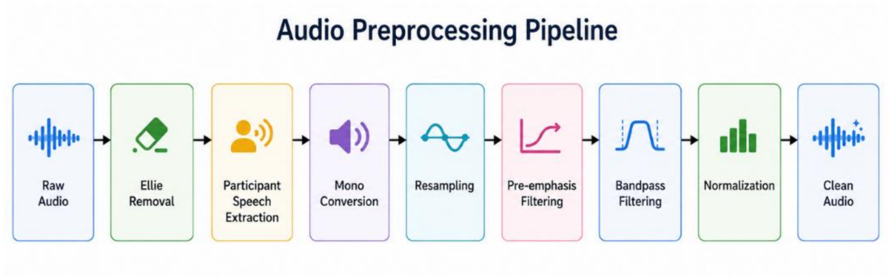

Fig. 2. Audio preprocessing pipeline: participant segment extraction, concatenation, mono conversion, pre-emphasis filtering, bandpass filtering, and normalisation.

3.3.2 Transcript Preprocessing.

Raw transcript files include XML-style tags (e.g., <laugh>), square-bracket markers, filler words and interviewer speech. The preprocessing pipeline consists of the following: (i) Remove any annotation tags from the utterances using regular expressions; (ii) Extract content from parenthetical expressions and (iii) Remove contentless utterances ‘uh’, ‘um’, ‘mm’, ‘hmm’,…; (iv) Keep only the content associated with a participant label; and (v) Normalise the whitespace. The cleaned transcripts yield good quality speech data for the participant in terms of linguistic features to be extracted. Transcript pre-processing pipeline is shown in figure 3.

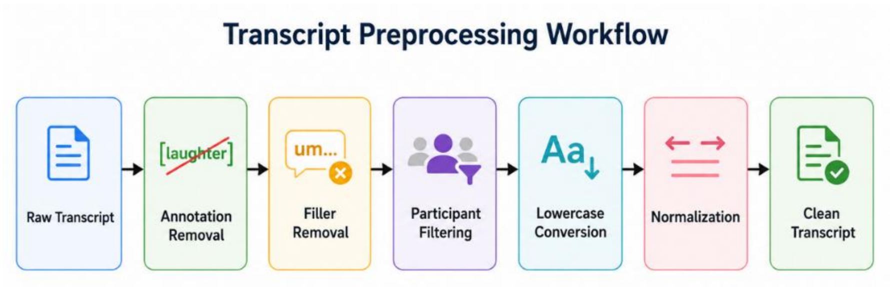

Fig. 3. Transcript preprocessing workflow: raw transcript ingestion, annotation removal, filler elimination, participant filtering, and output of cleaned CSV files.

3.3.3 Phase-wise Conversational Segmentation.

An important outcome of this study is the division of the clinical interview into three equally sized conversational segments. The sequential ordering of the participant's utterances is used to partition each cleaned transcript into three stages: Phase 1 (early interview), Phase 2 (mid interview), and Phase 3 (late interview. Audio segments are extracted from the audio file by mapping the phase assigned transcript timestamps back to the audio signal. This yields three different feature vectors for each participant, enabling classifiers to act on the temporally localised behavioural representations instead of on global interview statistics. The phase segmentation scheme is shown in figure 4.

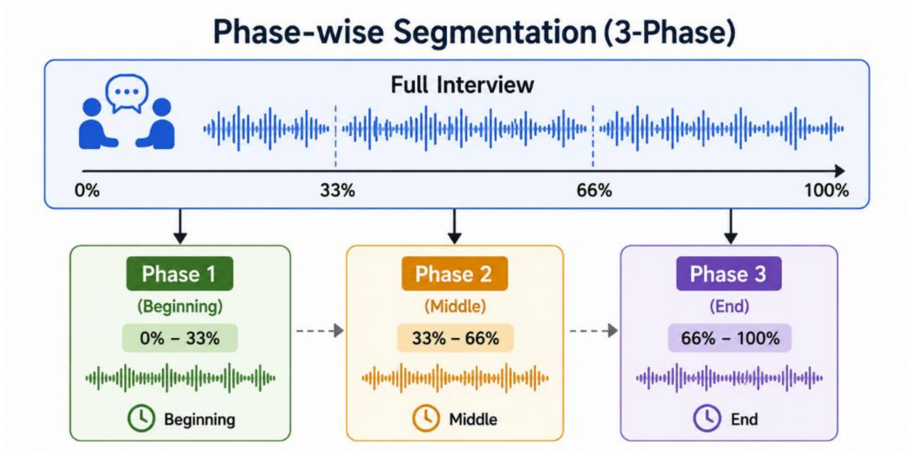

Fig. 4. Phase-wise conversational segmentation: each interview is divided into three equal phases based on transcript utterance ordering, with corresponding audio segments extracted using timestamp alignment.

3.3.4 Feature Extraction.

Feature extraction is done independently for each participant-phase pair for both modalities.

Librosa is used for acoustic feature extraction, and the wrapper to use librosa in conjunction with Parselmouth/Praat. Temporal features are speech duration, silence duration, speech-to-silence ratio, pause statistics (mean, SD, max, total).Energy features are speech duration, silence duration, speech-to-silence ratio, pause statistics (mean, SD, max, total). The spectral features include spectral centroid, bandwidth, rolloff, contrast and flatness. The 78 features are extracted by calculating the 13 MFCCs and associated delta coefficients. The voiced ratio is a pitch-related feature, along with the mean, standard deviation, range, and skewness. Voice quality measurements are obtained using the jitter, shimmer, and Harmonics to Noise ratio (local). Articulation speed is calculated as words per second of the time spent speaking, against the transcript. A total of 132 acoustic features are extracted for each participant-phase. Cleaned transcripts are used in the processes of linguistic feature extraction for each phase.

There are seven linguistic markers that are hand-crafted and computed: 1st person pronoun ratio, word count, sentence count, average sentence length, negation count, uncertainty word count, and type-token ratio. Also, a pause duration statistics (mean, max) is computed from transcript timestamps. The Sentence-BERT model all-MiniLM-L6-v2 is used to generate the semantic features, and it outputs 384-dimensional sentence embedding per utterance. There are 768 semantic features across the mean and the standard deviation, of the utterances within each phase. The text feature vector is a combination of the 777 dimensions. The multimodal feature architecture is shown in Figure 5.

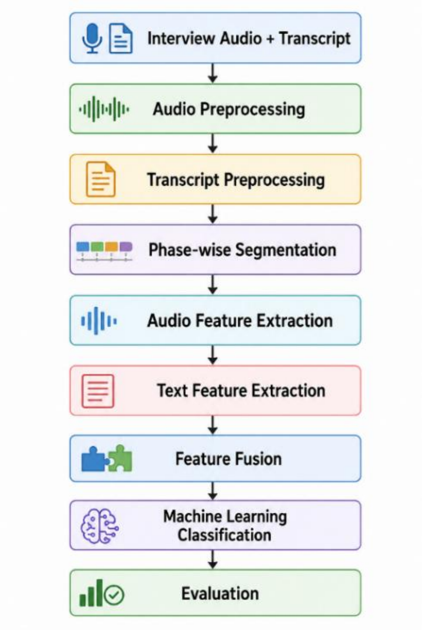

Fig. 5. Multimodal architecture: parallel acoustic and linguistic feature extraction pathways with feature-level fusion for classification.

Table 3. Summary of features extracted from audio and text modalities.

Modality	Category	Features Extracted
Audio	Temporal / Energy	Speech duration, silence duration, speech-to-silence ratio, silence ratio, speaking rate, pause mean/std/max/total, avg. segment duration
Audio	Prosodic	Pitch mean/std/min/max/range/median/skew, voiced ratio, jitter, shimmer, HNR mean/std
Audio	Spectral	Spectral centroid, bandwidth, rolloff, contrast, flatness (mean, std, var each)
Audio	Cepstral (MFCCs)	13 MFCCs × {mean, std, var} + 13 delta-MFCCs × {mean, std, var} = 78 features
Audio	Source/Energy	RMS mean/std, energy min/max/range/median/var, harmonic energy, percussive energy, onset count, articulation speed, word count
Text	Linguistic	Word count, sentence count, avg. sentence length, first-person ratio, negation count, uncertainty count, type-token ratio
Text	Temporal (Pauses)	Mean pause duration, max pause duration derived from transcript timestamps
Text	Semantic (SBERT)	384-dim sentence embedding mean + 384-dim std = 768 SBERT features per participant-phase
Text	Sentiment (VADER)	Positive, negative, neutral, compound sentiment scores (used in initial extraction)

3.3.5 Classification Models.

For the binary (depression/no depression) classification, six machine learning classifiers are used. CatBoost is a gradient boosting algorithm that provides native support for categorical features and has in-built regularisation for tabular data with mixed features. Random Forest is an ensemble of decision trees, which represents non-linear feature interactions and feature importance estimates. With strong performance on high-dimensional tabular data, XGBoost is a regularised gradient boosting with column subsampling. In Support Vector Machine (SVM) with radial basis function (RBF) kernel, features are mapped to high dimensional spaces for determining optimum class boundaries. Logistic Regression offers a linear baseline model, or baseline probability model. The Feedforward Neural Network with two hidden layers and ReLU activations learns complex non-linear patterns in the fused feature space.The Feedforward Neural Network with two hidden layers and ReLU activations learns complex non-linear patterns in the fused feature space.

4. Experimentation and Model Evaluation

4.1 Experimental Setup

The experiments are all run on a desktop computer with an Intel Core i7 12650H processor and 16GB of RAM running on Windows 11. The software environment contains Python 3.12, librosa 0.10, Parselmouth 0.4, scikit-learn 1.4, XGBoost 2.0, CatBoost 1.2, sentence-transformers 2.7 and PyTorch 2.1. Feature extraction of audio files was carried out using the directory of WAV files isolated for each participant, called cleaned_audio, and the transcription process was done in Google Colab.

Three different evaluation setups are used: (i) one train-test split (80:20), (ii) 5-fold stratified cross validation, and (iii) 10-fold stratified cross validation. The class distribution is maintained through stratification. Standard metrics are computed for each configuration:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

$$Precision = TP / (TP + FP); \quad Recall = TP / (TP + FN)$$

$$F1\text{-}Score = 2 \times (Precision \times Recall) / (Precision + Recall)$$

All feature matrices were scaled before classification using the StandardScaler. The Feedforward Neural Network consists of 2 hidden layers of 256 and 128 units each trained with ReLU activation and dropout (p = 0.3); this is done for 100 epochs using the Adam optimiser with a learning rate of 0.001.

4.2 Results and Outputs

4.2.1 Dataset Distribution.

The DAIC-WOZ data set is imbalanced with 297 (70.2%) samples labeled as non-depressed and 126 (29.8%) labeled as depressed in total 423 samples across the participant-phase. There are 141 samples in each of the three phases, and the class proportions are the same throughout the phases, indicating consistent stratification. The data set class distribution is shown in Figure 6.

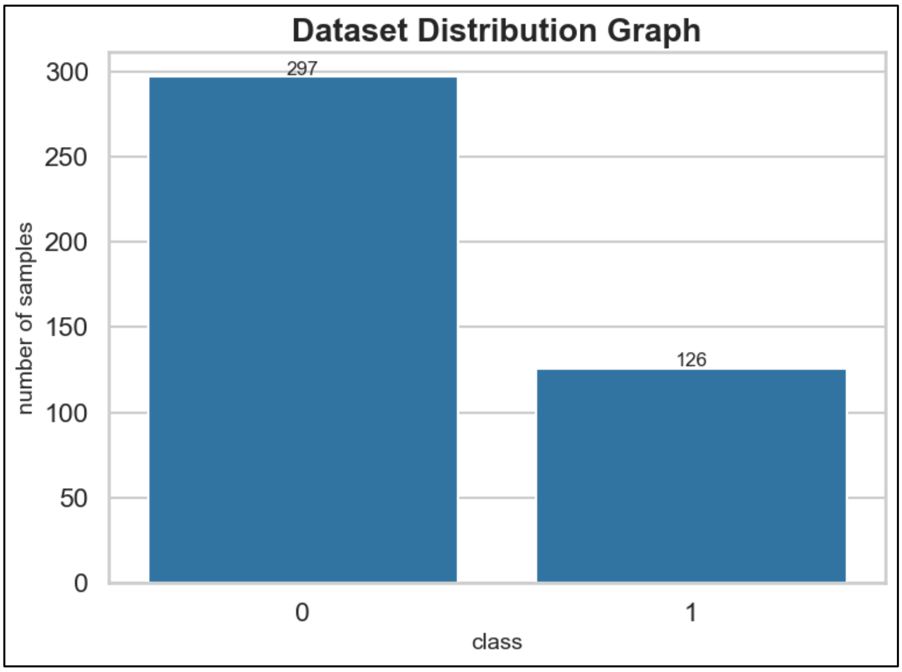

Fig. 6. Class distribution of the DAIC-WOZ dataset across phases: non-depressed (0) versus depressed (1) sample counts per phase.

4.2.2 Model Performance Comparison.

The results of all six classifiers with 10-fold cross validation in Phase 1 with Text-only, Audio-only and multimodal feature configuration are shown in Table 4. Among all the models, the multimodal configuration yields the best accuracy and overall F1-scores. Random Forest got the best overall performance with 83.6% accuracy and F1-score of 0.71, followed by XGBoost (82.3%, F1: 0.67) and SVM (82.5%, F1: 0.65). CatBoost is the most accurate model in the text only configuration (83.9 % accuracy) with lower recall for the depressed class. A model that only uses acoustic information performs significantly worse, thus supporting the idea that acoustical information alone is not enough for reliable depression classification in this dataset.

Table 4. Model performance under 10-fold cross-validation, Phase 1.

Multimodal Scores				
Model	Accuracy	Precision	Recall	F1-Score
Random Forest 10 Folds	83.60%	0.6471	0.7857	0.7097
XGBoost 10 Folds	82.32%	0.5556	0.8333	0.6667
SVM 10 Folds	82.49%	0.617	0.6905	0.6517
Logistic Regression 10 Folds	82.97%	0.625	0.7143	0.6667
CatBoost 10 Folds	80.11%	0.5536	0.7381	0.6327

Text-Only				
Models	Accuracy	Precision	Recall	F1-Score
Random Forest 10 Folds	74.54%	0.6667	0.1905	0.2963
XGBoost 10 Folds	68.59%	0.4528	0.5714	0.5053
SVM 10 Folds	54.35%	0.3143	0.2619	0.2857
Logistic Regression 10 Folds	61.78%	0.3241	0.8333	0.4667
CatBoost 10 Folds	83.91%	0.6458	0.7381	0.6889

Audio-Only				
Models	Accuracy	Precision	Recall	F1-Score
Random Forest 10 Folds	67.51%	0.52	0.3095	0.3881
XGBoost 10 Folds	66.31%	0.3774	0.4762	0.4211
SVM 10 Folds	61.50%	0.4286	0.0714	0.1224
Logistic Regression 10 Folds	60.65%	0.359	0.6667	0.4667
CatBoost 10 Folds	63.23%	0.4375	0.5	0.4667

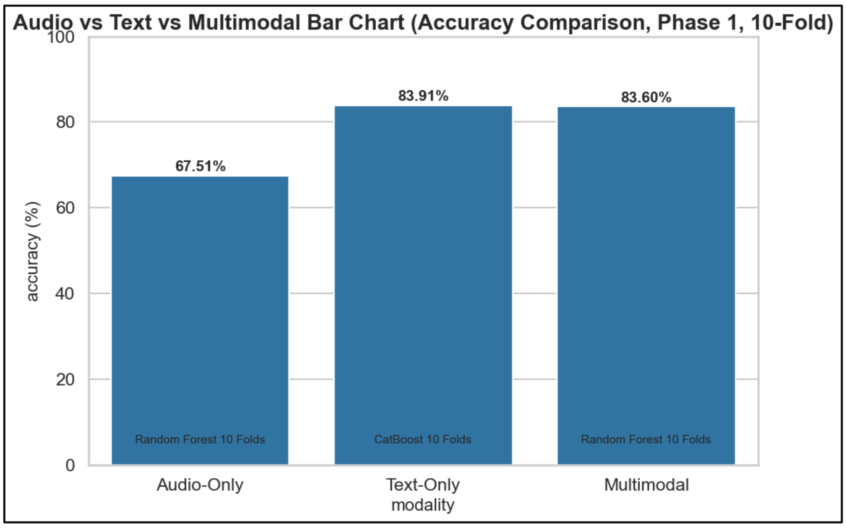

Fig. 7. Comparative accuracy of audio-only, text-only, and multimodal configurations across classifiers for Phase 1 under 10-fold cross-validation.

4.2.3 Phase-wise Analysis.

The multimodal performance of all classifiers is given in Table 5 under 10-fold Cross-Validation for the three conversational phases. An important and consistent pattern emerges that in all the models, the accuracy and F1 scores are highest for Phase 1, showing that the early responses in the interview have the largest discriminative depression signals. The performance deteriorates gradually in Phase 2 and Phase 3 and is the worst in Phase 3. Similar results are seen for all classifiers with random forest obtaining 83.6%, 79.3% and 77.9% accuracy in Phases 1, 2 and 3 respectively.

Table 5. Phase-wise model performance under multimodal 10-fold cross-validation.

Multimodal												
Model	Phase 1				Phase 2				Phase 3			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Random Forest 10 Folds	83.60%	0.6471	0.7857	0.7097	79.29%	0.6316	0.5714	0.6	77.85%	0.6562	0.5	0.5676
XGBoost 10 Folds	82.32%	0.5556	0.8333	0.6667	75.88%	0.4808	0.5952	0.5319	71.69%	0.4516	0.6667	0.5385
SVM 10 Folds	82.49%	0.617	0.6905	0.6517	76.62%	0.5208	0.5952	0.5952	75.76%	0.5778	0.619	0.5977
Logistic Regression 10 Folds	82.97%	0.625	0.7143	0.6667	73.02%	0.5	0.619	0.5532	70.95%	0.4902	0.5952	0.5376
CatBoost 10 Folds	80.11%	0.5536	0.7381	0.6327	76.98%	0.551	0.6429	0.5934	66.35%	0.4694	0.5476	0.5055

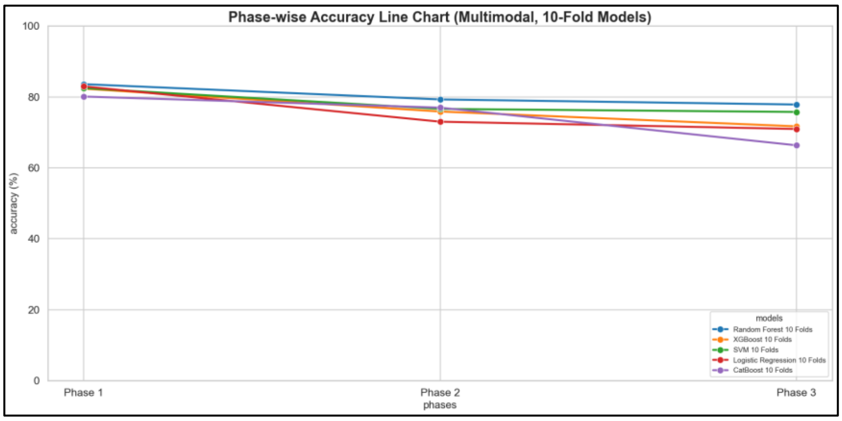

Fig. 8. Phase-wise accuracy trends for all classifiers under multimodal 10-fold cross-validation, illustrating progressive performance decline across conversational phases.

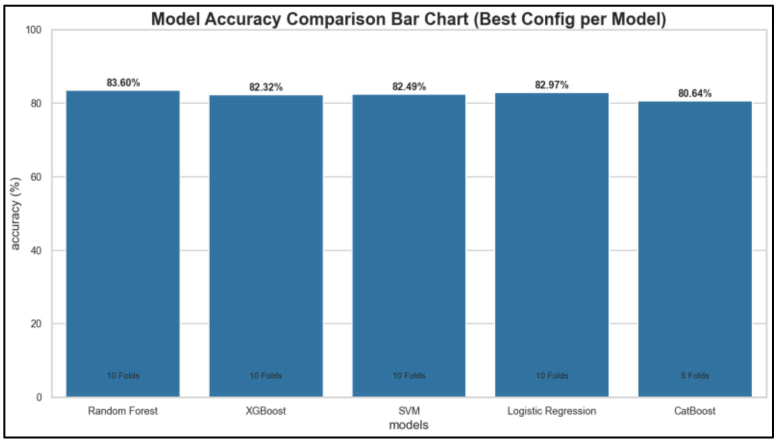

Fig. 9. Comparative model accuracy under the best-performing configuration (multimodal, 10-fold cross-validation, Phase1).

4.2.4 Feature Importance Analysis.

We conducted an analysis of which features are important. We applied the Random Forest classifier model to the features that we obtained from Phase 1. The results indicate that there are some features that are indeed very valuable for the task of classification from the SBERT embeddings. The features sbert_std_56, sbert_std_5 and sbert_mean_7 are the important ones. We also saw that some acoustic and speech features, like pause_std, pause_mean MFCC-1 mean and spectral_contrast_std are important too.The SBERT displays the mean and standard deviation are of paramount importance. This demonstrates that the meanings of words and their sound patterns are significant. All of the features in the SBERT are complemented by the acoustic features, which all help us to understand what's happening. Still very important features of the

SBERT are like sbert_std_56 and sbert_mean_7. The most important 20 features are given in figure 10.

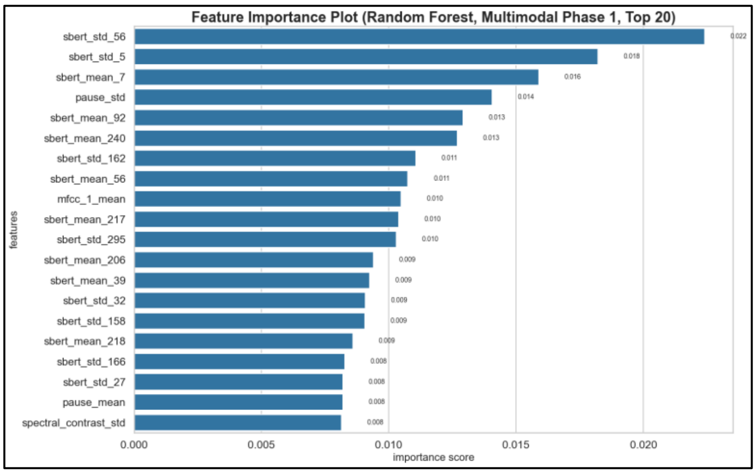

Fig. 10. Top 20 most important features from the multimodal Random Forest model (Phase 1, 10-fold cross-validation), spanning acoustic, prosodic, and semantic modalities.

4.2.5 Confusion Matrix.

Figure 11 shows us the confusion matrix for the multimodal Random Forest model. This is the model uses 10 fold cross-validation on the Phase 1 data. The confusion matrix will give us an idea that the multimodal Random Forest is the best model among the other models selected for classification. It doesn't make mistakes when it says that something isn't depressed when it is. This is very important when screening, as missing a depressed individual is worse than saying that a person is depressed when they aren't. This is being accomplished by the multimodal Random Forest model. The validation / system performance evaluation was performed.Validation / System Performance Evaluation was conducted.

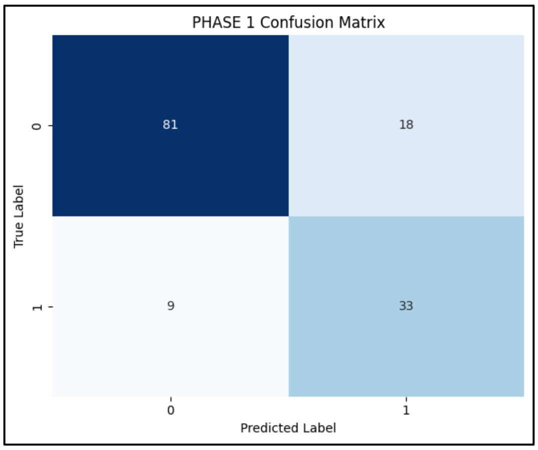

Fig. 11. Confusion matrix for the best-performing multimodal Random Forest classifier (Phase 1, 10-fold cross-validation).

5. Validation / System Performance Evaluation

From the experimental results several significant analytical observations are made about the effectiveness of the proposed system. However, the clear superiority of the multimodal configuration over both unimodal configurations is consistent with the fact that acoustic and linguistic modalities provide complementary and, to a large extent, non-redundant information about depressive states. Compared with the multimodal models, the models based on only text have similar overall performance in terms of absolute accuracy (CatBoost text-only: 83.9% vs. Random Forest multimodal: 83.6% in Phase 1), but produce more balanced precision-recall curves, which has more clinical relevance in screening applications.

It is notable that the Phase 1 performance is so much better than later phases. The most conducive interview segments seem to be those in which the participants are answering introductory and self-referential questions, where the richest indicators of depression are present. This is in line with clinical experience that depressed patients use limited words, more first-person singular pronouns and modified prosodic patterns while talking about their feelings. In Phases 2 and 3 of the interview these unique patterns of behaviour might become less common or more inconsistent as the interview becomes more topic-specific and structured, making classifiers less likely to distinguish the speaker.

The good performance on text only features is due to the regularised gradient boosting mechanism used by CatBoost, which works well with the high dimensional (777-features) SBERT dominated feature space. On the contrary, the superiority of the Random Forest model in the multimodal configuration indicates that ensemble decision-tree models work well with the heterogeneous space of acoustic-linguistic features. The Feedforward Neural Network performed generally poorly, although it had very high recall for the depressed class, this suggests that in this regime of data size there is a high false positive rate, which is consistent with smaller data sets being required for deep learning to be reliable across generalisation.

All the audio-only models and phases perform significantly worse than the audio-visual-only models, revealing the shortcomings of audio-only features as a standalone element. The high precision but low recall of audio-only Random Forest (e.g., 52.0% precision, 31.0% recall, Phase 1) suggests that there might be some acoustic patterns that are specific to depression but which alone are not enough to cover the range of depressive behavioural markers when using only the auditory information.

The preprocessing pipeline is reflected in the quality of the features which the pipeline produces: Using ground-truth timestamps, participant-only audio has been extracted, thus directly affecting the reliability of features based on prosody and energies. Likewise, transcript cleaning means that the semantic embedding is an accurate representation of the language used by participants and not of annotations that are artifacts.

6. Discussions on Contributions

This study makes a number of methodologically unique contributions to the field of computational mental health analysis. The first is the design philosophy of the preprocessing; it is a philosophy that puts the focus of the data preparation on the isolation of the participants, spectral filtering and transcript cleaning as primary goals of data preparation and not as afterthoughts necessary to achieve optimal features for downstream analyses.The first is the design philosophy of the preprocessing; in this design philosophy, primary goals of the data preparation are to isolate the data of the participants, spectral filtering and cleaning of the transcripts, and these are not tasks that are carried out once the features are extracted, in order to obtain good ones for downstream analysis.

Secondly, the phase-wise conversational segmentation is a novel analytical dimension that goes beyond the current paradigm of whole interview analysis. This research also shows that features from Phase 1 are systematically more discriminative than features from later phases, which means that there is actionable knowledge in the design of a shorter, more targeted screening instrument that targets early interview responses.

Then, the multimodal representation of the acoustic features, which was built using an extensive feature engineering process (132 acoustic features such as MFCCs with delta coefficients, along with voice quality measures (jitter, shimmer, HNR)), along with a full set of the spectral descriptors, and the text features (777 text features, including Sentence-BERT semantic embeddings and linguistic markers), results in a rich, interpretable 909-dimensional representation that does not rely on black-box end-to-end deep learning architectures.

At last, the systematic, comparative evaluations are carried out over six classifiers, three evaluation protocols, three conversational phases and three modality configurations, resulting in a detailed performance panorama that can give sound empirical directions to choose models for clinical NLP tasks. The comparison results of the models, which show that CatBoost is the strongest text-only model and Random Forest is the strongest multimodal model, practically give them a hint for action.

7. Conclusion and Future Scope

In this paper, a multimodal framework for depression detection based on clinical interview data was introduced, focusing on the preprocessing stage. The system extracted 132 acoustic and 777 linguistic features per participant-phase record, tested six machine learning classifiers for each of three conversational phases and multiple cross validation procedures on the DAIC-WOZ corpus. Key findings include: (i) baseline unimodal models are outperformed by the multimodal configuration, which achieves 83.6% accuracy and 0.71 F1-score with Random Forest under 10-fold cross-validation; (ii) the accuracy of the models is highest in Phase 1, suggesting that early responses in interviews contain the richest information about depression behaviour; and (iii) CatBoost and Random Forest are the most reliable classifiers for this task, while deep learning models require larger datasets to fully realize their potential.

There are some restrictions to be noted. Although widely used, the DAIC-WOZ data set includes 141 participants which might constrain generalization, especially for deep learning models. The class imbalance (70:30) can cause a bias of classifiers toward the majority class. Pause types based on transcript timestamps may not be exactly accurate representations of the actual pause distributions. These limitations will be covered in future work by: (i) adopting transformer-based language models (e.g., RoBERTa, BioMedBERT) to obtain more detailed semantic feature extraction; (ii) examining attention-based multimodal fusion architectures; (iii) expanding to larger and more demographically diverse corpora; (iv) incorporating real-time audio processing for clinical screening applications; and (v) studying cross-corpus generalisation to evaluate robustness across various interview protocols and recording conditions.

References

[1] Y. Guo, C. Zhu, S. Hao, and R. Hong, “A topic-attentive transformer-based model for multimodal depression detection,” arXiv preprint arXiv:2206.13256, 2022.

[2] L. Gan, Y. Huang, X. Gao, J. Tan, F. Zhao, and T. Yang, “Multimodal magic: Elevating depression detection with a fusion of text and audio intelligence,” Preprint, 2025.

[3] Y. Gong and C. Poellabauer, “Topic modeling based multi-modal depression detection,” in Proc. 7th Annu. Workshop Audio/Visual Emotion Challenge (AVEC ’17), ACM Multimedia, 2017.

[4] M. Nykoniuk, O. Basystiuk, N. Shakhovska, and N. Melnykova, “Multimodal data fusion for depression detection approach,” Computation, vol. 13, no. 1, p. 9, 2025.

[5] H. Yoo and H. Oh, “Depression detection model using multimodal deep learning,” Preprints, 2023.

[6] Z. Xu, Y. Gao, F. Wang, L. Zhang, L. Zhang, J. Wang, and J. Shu, “Depression detection methods based on multimodal fusion of voice and text,” Scientific Reports, vol. 15, p. 21907, 2025.

[7] J. Park and N. Moon, “Design and implementation of attention depression detection model based on multimodal analysis,” Sustainability, vol. 14, no. 6, p. 3569, 2022.